

AI Assisted Coding

Assignment – 1.5

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Batch: 24

Task – 1 Prompt : Use GitHub Copilot to generate a Java program that reverses a user-input string using inline logic without functions.

Code :

```
import java.util.Scanner;

public class ReverseString {

    public static void main(String[] args) { Scanner
        scanner = new Scanner(System.in);
        System.out.print("Enter a string: ");
        String input = scanner.nextLine();
        String reversed = "";
        for (int i = input.length() - 1; i >= 0; i--) { reversed +=
            input.charAt(i);
        }
        System.out.println("Reversed string: " + reversed);
    }
}
```

Output :

Enter a string: SR University

Reversed string: ytisrevinU RS

Task – 2 Prompt : Use GitHub Copilot to optimize the string reversal code by simplifying logic, reducing variables, and improving readability, and explain how the changes improve efficiency.

Code :

```
import java.util.Scanner;

public class ReverseString{

    public static void main(String[] args) { Scanner
        scanner = new Scanner(System.in);
        System.out.print("Enter a string: ");
        String input = scanner.nextLine(); StringBuilder
        reversed = new StringBuilder(); for (int i =
        input.length() - 1; i >= 0; i--) {
            reversed.append(input.charAt(i));
        }
        System.out.println("Reversed string: " + reversed.toString());
    }
}
```

Output :

```
Enter a string: SR University
Reversed string: ytisrevinU RS
```

Task – 3 Prompt : Use GitHub Copilot to generate a function-based Java program that reverses a string, returns the result, and includes meaningful comments.

Code :

```
import java.util.Scanner;

public class ReverseString {

    public static String reverseString(String input) {
        StringBuilder reversed = new StringBuilder(); for
        (int i = input.length() - 1; i >= 0; i--) {
            reversed.append(input.charAt(i));
        }
        return reversed.toString();
    }
}
```

```

public static void main(String[] args) { Scanner
    scanner = new Scanner(System.in);
    System.out.print("Enter a string: ");
    String input = scanner.nextLine(); String
    reversed = reverseString(input);
    System.out.println("Reversed string: " + reversed);
}
}

```

Output :

Enter a string: AI Assisted Coding
 Reversed string: gnidoC detsissA IA

Task – 4 Prompt : Use GitHub Copilot to compare procedural and modular string reversal programs and analyze them based on clarity, reusability, debugging, and scalability.

Code :

```

import java.util.Scanner;

public class ReverseString {
    // Function to reverse a given string
    public static String reverseString(String input) {
        StringBuilder reversed = new StringBuilder(); for
        (int i = input.length() - 1; i >= 0; i--) {
            reversed.append(input.charAt(i));
        }
        return reversed.toString();
    }
}

```

```

public static void main(String[] args) { Scanner
    scanner = new Scanner(System.in);
    System.out.print("Enter a string: ");
    String input = scanner.nextLine();
    // Call the reverseString function and store the result String
    reversed = reverseString(input); System.out.println("Reversed
    string: " + reversed);
}
}

```

Output :

Enter a string: Samshray

Reversed string: yarhsmaS

Task – 5 Prompt : Use GitHub Copilot to generate loop-based and slicing-based string reversal methods and compare them based on execution flow, time complexity, performance, and suitability.

Code :

```

import java.util.Scanner;

public class ReverseString {
    // Loop-based method to reverse a string
    public static String reverseStringLoop(String input) { StringBuilder
        reversed = new StringBuilder();
        for (int i = input.length() - 1; i >= 0; i--) {
            reversed.append(input.charAt(i));
        }
        return reversed.toString();
    }

    // Slicing-based method to reverse a string

```

```

public static String reverseStringSlicing(String input) { return new
    StringBuilder(input).reverse().toString();
}

public static void main(String[] args) { Scanner
    scanner = new Scanner(System.in);
    System.out.print("Enter a string: ");
    String input = scanner.nextLine();

    // Using loop-based method
    String reversedLoop = reverseStringLoop(input); System.out.println("Reversed string
(loop-based): " + reversedLoop);

    // Using slicing-based method
    String reversedSlicing = reverseStringSlicing(input); System.out.println("Reversed string
(slicing-based): " + reversedSlicing);
}
}

```

Output :

Enter a string: SR University

Reversed string (loop-based): ytisrevinU RS

Reversed string (slicing-based): ytisrevinU RS